MAT367
Intro to Differential Geometry

Class Lecture Notes

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Class Lectures

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

VII Stokes and Poincare

15 Stokes' Theorem

Theorem 15.1: Generalized Stokes' Theorem

let $A \subseteq M^m$ be a *bounded region* with *boundary* ∂A. then for any $\alpha \in \Omega^{m-1}(M)$ with $\operatorname{supp}(\alpha) \cap A$ cpt, we have:

$$\int_{\partial A} \omega = \int_A \mathrm{d}\omega$$

remark: M may be a submfld of sth higher-dim! other remark: we literally havent talked abt bdries lmao

Example 15.2

i) sps $\alpha \in \Omega^2(\mathbb{R}^3)$, A some bounded region in $\mathbb{R}^3$. then this yields the divergence theorem.

ii) sps $\alpha \in \Omega^1(\mathbb{R}^3)$, A some bounded surface in $\mathbb{R}^3$. then this yields the classical stokes thm.

iii) sps $\alpha \in \Omega^0(M) = C^\infty(M)$. then this yields the fundamental thm of calculus.

gave 2 exs w pictures. oh well

Corollary 15.3

(more of a special case) if α closed, then:

$$\int_{\partial A} \alpha = 0$$

Definition 15.4

sps M oriented. we say $D \subseteq M$ is a **region with smooth boundary** if $\exists f : C^\infty(M)$ with 0 as a reg val and $D = f^{-1}(t \leq 0)$. we write $\partial D = f^{-1}(0)$ and $\text{int}(D) = f^{-1}(t < 0)$.

fix a special atlas covering D:

- cvr ∂D by submfld charts (U, φ) compat w orientation of M, ie:

$$\varphi(U \cap \partial D) = \varphi(U) \cap \{x_1 = 0\} \qquad \varphi(U \cap \text{int}D) = \varphi(U) \cap \{x_1 < 0\}$$

  after possibly flipping sign of x_1

- cvr interior by more oriented charts

this gives an oriented atlas on D.

Proposition 15.5

when restricting to $\{x_1 = 0\}$, we get an oriented atlas on ∂D

Proof.

let $\tau : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j)$ be the transition map. given $x = (0, x_2, \ldots, x_m)$, consider $T_x\tau$: the top left is > 0 since "away from D" is preserved, the rest of the first row is 0 [cz bdry is preserved? idk], and ignoring the first column, the remaining matrix is made up of the jacobian of the transition map btwn charts on ∂D.

we know $\det T_x\tau > 0$, so det(that other thing) is +ve, so the atlas $\left(U\big|_{\partial D}, \varphi\big|_{\partial D}\right)$ is oriented. ■

note: the orientation we put on ∂D is engineered to avoid a minus sign.

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today, we prove stokes.

recall we can pick oriented atlas on D which induces an orientation on ∂D. since α cpt supp on D, its sppted on finitely many charts. pick a partition of unity ρ_i sppted on:

$$\{U_i\} \cup (M \setminus D \cap \text{supp}(\alpha))$$

then, we can write:

$$\alpha \;=\; \sum_i \rho_i \alpha \qquad \rho_i \alpha \text{ supp on } U_i$$

thus, its enough to show stokes for α cpt supp on $\mathbb{R}^m$, $D = \{x : x_1 \leq 0\}$. then:

$$\alpha \;=\; \sum_{i=1}^{m} f_i \; \mathrm{d}x_1 \wedge \cdots \wedge \widehat{\mathrm{d}x_i} \wedge \cdots \wedge \mathrm{d}x_m$$

evaluating at $(\partial_2, \ldots, \partial_m)$, we get:

$$\int_{\{x_1=0\}} \alpha \;=\; \int_{\mathbb{R}^{m-1}} \iota^* \alpha \;=\; \int_{\mathbb{R}^{m-1}} f_1 \; \mathrm{d}x_2 \wedge \cdots \wedge \mathrm{d}x_m$$

otoh:

$$\mathrm{d}\alpha \;=\; \left[\sum_{i=1}^{m} (-1)^{i+1} \partial_i f_i \right] \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_m$$

so:

$$\int_{\{x_1 \leq 0\}} \mathrm{d}\alpha \;=\; \sum_{i=1}^{m} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \int_{-\infty}^{0} \partial_i f_i \; \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_m$$

note that when $i \neq 1$, this is equal to 0. indeed, we can integrate $\mathrm{d}x_i$ first via fubinis, then by ftc:

$$\int_{-\infty}^{\infty} \partial_i f_i \;=\; \lim_{x \to \infty} f_i - \lim_{x \to -\infty} f_i \;=\; 0$$

when $i = 1$, we get:

$$\int_{\mathbb{R}^{m-1}} \left[\int_{-\infty}^{0} \partial_1 f_1 \; \mathrm{d}x_1 \right] \;=\; \int_{\mathbb{R}^{m-1}} f_1(0, x_2, \ldots, x_m) \; \mathrm{d}x_2 \cdots \mathrm{d}x_m \;=\; 0$$

Corollary 15.6

if M^m orientable, $\alpha \in \Omega^m(M)$ (w cpt sppt) is exact, ie $\alpha = \mathrm{d}\omega$, and ω cpt sppt, then $\int_M \alpha = 0$

fact: if M is connected, the converse is also true.

Corollary 15.7

if M orientable cpt conn, then $H^m(M) \simeq \mathbb{R}$, where H^m is the m-th de Rham cohomology group.

if M not cpt, then $H_c^m(M) \simeq \mathbb{R}$, where we restrict to forms w cpt sppt.

Proposition 15.8

every orient M admits a *volume form*, ie an m-form vanishing nowhere

note: if M cpt, $\int_M \omega \neq 0$. otw, may not have finite integral.

Proof.

take locally finite oriented atlas, partition of unity ρ_i, and:

$$\omega = \sum_i \varphi_i^*(\rho_i \circ \varphi_i^{-1})\mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_m$$

note each summand extends over M since its 0 outside cpt subset of U_i. an orientation ensures signs are all the same, so $\omega \neq 0$. ■

Proposition 15.9

if M admits a vol form, its orientable

Proof.

at each pt, given basis $X_1, \ldots, X_m$ of T_pM, $\omega_p(X_1, \ldots, X_m) > 0$ or < 0. this defns an orientation at a pt; check that this defns a global orientation [see hw]. ■

Theorem 15.10

sps $\omega \in \Omega^k(M)$ closed, $F : \mathbb{R} \times S \longrightarrow M$ smooth, considered as a family of smooth $F_t : S \longrightarrow M$. then $\int_S F_t^* \omega$ is indep of t.

Proof.

consider $D = [a, b] \times S$:

$$0 = \int_D F^* \mathrm{d}\omega = \int_D \mathrm{d}F^* \omega = \int_{\partial D} F^* \omega = \int_S F_b^* \omega - \int_S F_a^* \omega$$

■

16 Poincare(-Hopf) Theorem

we know what winding number is. note that:

1) $\gamma(t) = (\cos(2\pi nt), \sin(2\pi nt))$ has winding num n

2) winding num is inv under deformations that stay in the punctured plane.

number 2 suggests we can define $\int_\gamma$(closed form).

Definition 16.1

the **winding number** of a loop γ is given by:

$$w(\gamma) \;=\; \int_{S^1} \gamma^* \left(\frac{\mathrm{d}\theta}{2\pi}\right)$$

note θ is as in polar coords, but θ is not a smooth fn $\mathbb{R}^2 \setminus \{0\} \longrightarrow \mathbb{R}$, so $\mathrm{d}\theta$ is not exact.

compute: if $x = r\cos\theta, r^2 = x^2 + y^2$, then:

$$\mathrm{d}x \;=\; -r\sin\theta \mathrm{d}\theta + \mathrm{d}r\cos\theta \qquad 2r\mathrm{d}r \;=\; 2x\mathrm{d}x + 2y\mathrm{d}y$$

therefore:

$$\mathrm{d}\theta \;=\; \frac{x\mathrm{d}y - y\mathrm{d}x}{x^2 + y^2}$$

check its closed. this satisfies 1) by computation, and 2) by properties of closed forms.

moreover:

1) $w(\gamma) \in \mathbb{Z}$
2) two loops can be deformed into each other iff they have the same winding num
3) joining loops by a path results in a loop whose winding num is the sum of the others' winding nums

Theorem 16.2: Poincare's Theorem

let Σ be cpt oriented surface w ∂, X VF on Σ w *isolated zeroes* (or equivalently by cptness, finitely many). let $\mathcal{C}$ denote the connected components of the bdry (ie circles). then:

$$\sum_{0\text{'s}} \mathrm{index}_p(X) - \sum_{C \in \mathcal{C}} \mathrm{rot}_C(X) \;=\; \chi(\Sigma)$$

where χ is the euler characteristic.

heuristic: $\forall X$ VF, sth abt how it behaves near 0's + sth abt how it behaves at $\partial\Sigma =$ topological invariant.

the euler characteristic has many defns. for instance:

$$\chi(M) \;=\; \sum_{i=0}^{m}(-1)^i \dim(H^i(M))$$

we take an axiomatic route.

Definition 16.3

the euler characteristic is a fn (for cpt surfaces) that satisfies:

E1) if $\Sigma = \Sigma' \sqcup \Sigma''$, then $\chi(\Sigma) = \chi(\Sigma') + \chi(\Sigma'')$

E2) if Σ obtained from Σ' by gluing two ∂ circles, then $\chi(\Sigma) = \chi(\Sigma')$

E3) $\chi(D^2) = 1$

Example 16.4

since $S^2 = D^2 \cup_\partial D^2$, $\chi(S^2) = 2$.

Corollary 16.5: Hairy Ball Theorem

every VF on S^2 has a zero

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fact that was forgotten last time: given a cpt orient surface w genus g and r bdry circles (think sliced torus), $\chi(\Sigma) = 2 - 2g - r$.

consider first a VF on D^2, $X_p = f(p)\partial_x + g(p)\partial_y$. we identify it as $F_X = (f, g) : D^2 \longrightarrow \mathbb{R}^2$. if γ a loop on D^2 st $X_p \neq 0$ on γ, then we can define:

$$\mathrm{rot}_\gamma(X) \;=\; w(F_X \circ \gamma)$$

note that if γ goes through a zero, it can be deformed arnd it such that the rot stays the same (by stokes).

given an isolated 0 p of X, then any curve γ in nonzero nbhd B of p w winding num 1 arnd p has the same rot num.

Definition 16.6

$\text{index}_p(X) = \text{rot}_\gamma(X)$ for any such γ as above.

Exercise 16.7

this doesn't depend on choice of chart

next, we consider rot num along a bdry circle. note that we can always get a VF X_1 tangent to a circle C, and if C is a ∂ circle, we can defn X_2 as facing "inward". then any other VF X along C can be written as:

$$X_p \;=\; f(p)(X_1)_p + g(p)(X_2)_p$$

so we can define:

$$\text{rot}_C(X) \;=\; w(F_X \circ \gamma)$$

where γ is a parametrization of C, and $F_X = (f, g) : C \longrightarrow \mathbb{R}^2$ as before.

proof sketch of poincaré: general strat is to prove LHS satisfies axioms for χ, then invoke uniqueness.

E1) get this man a true (i wasnt paying attn)

E2) X agrees on the two circles since "inward" vec fields are opposite, so rot nums cancel

E3) by additivity of winding num, we can join loops arnd zeroes to get a curve γ, so $\text{rot}_\gamma(X) = \sum_p \text{index}_p(X) = \text{rot}_{\partial D^2}(X) = \text{rot}_C(X) - 1$ wrt the usual basis, thus LHS is 1.

and thats all folks!